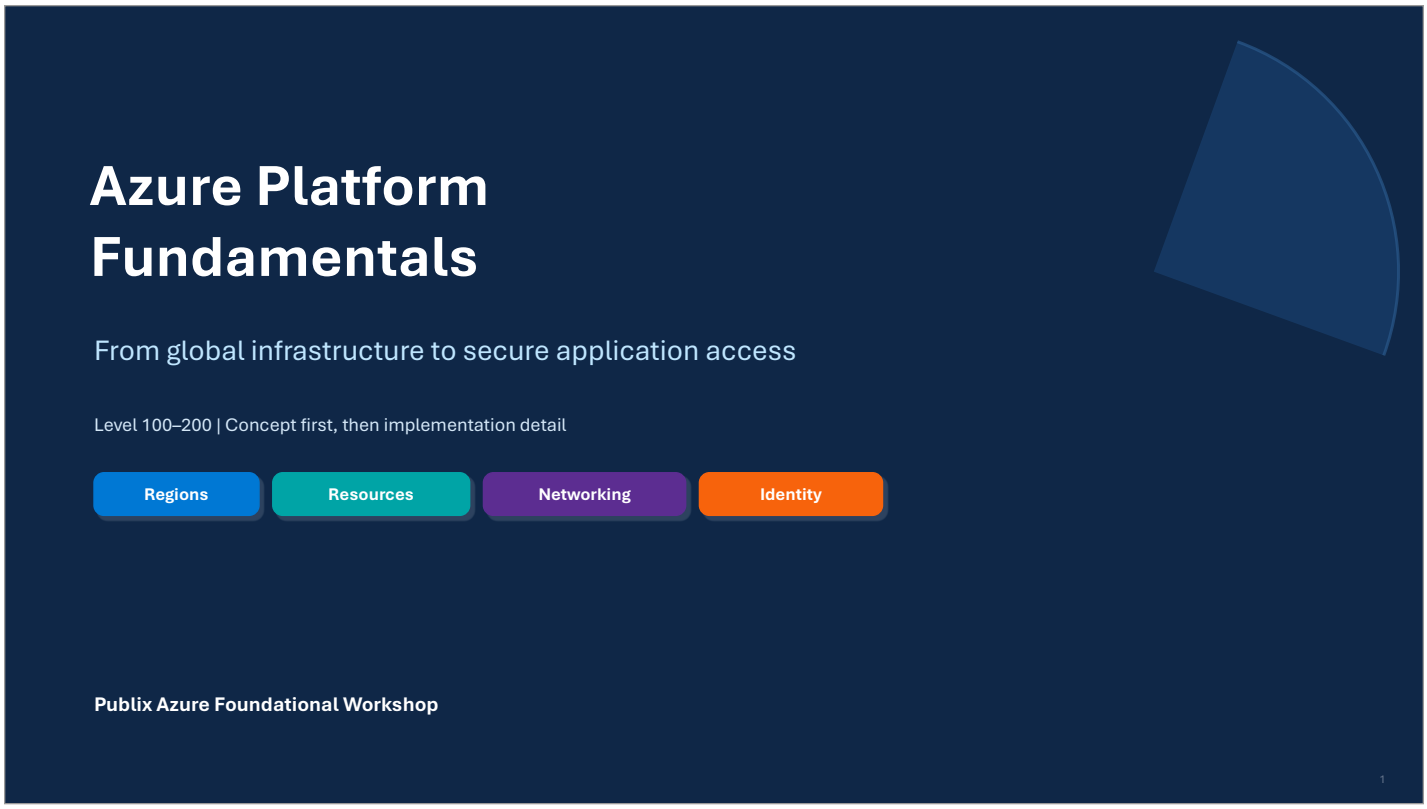
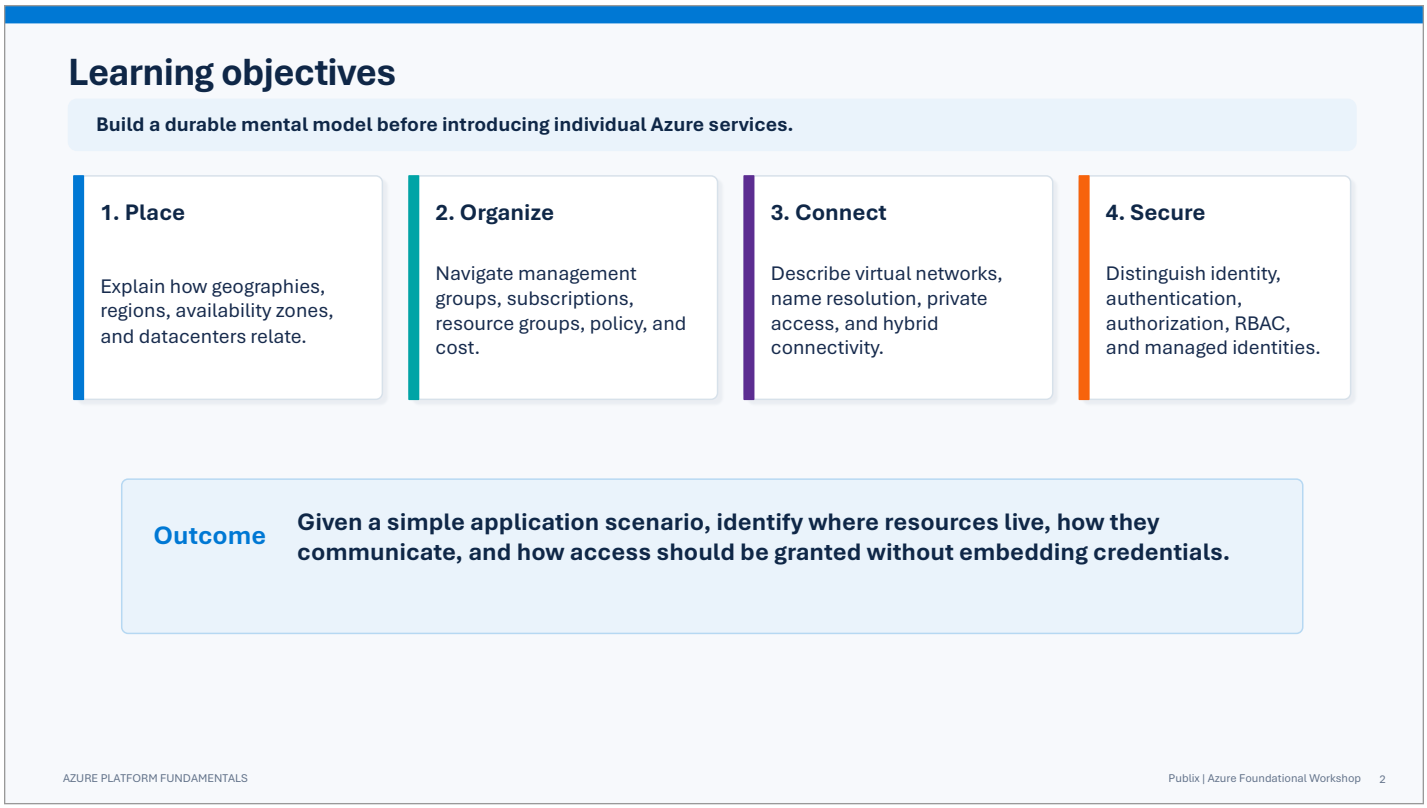


Slide 1 of 21

Opening: This module establishes the platform concepts that make later application, integration, and operations topics easier to understand. Emphasize the progression from physical infrastructure, to management scope, to connectivity, to identity and authorization.

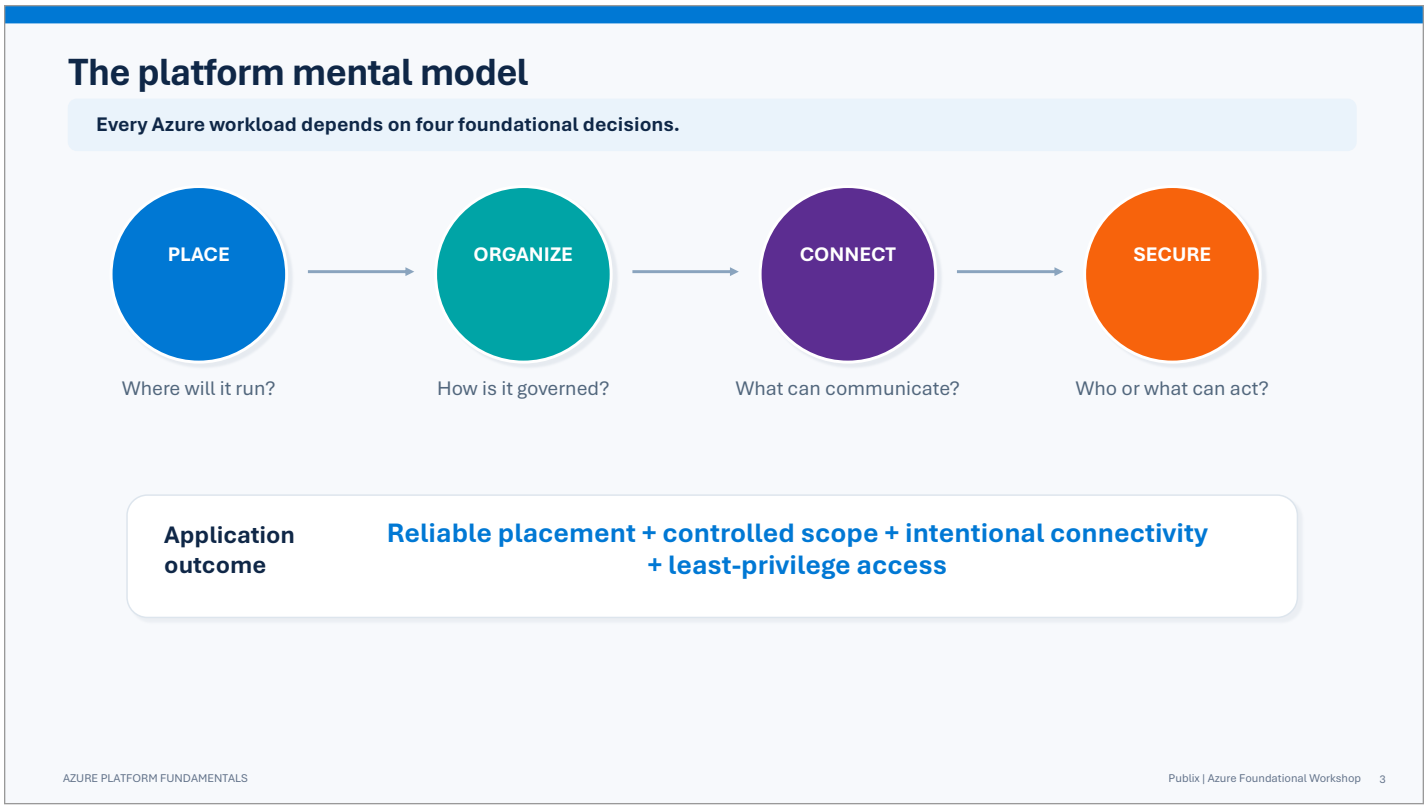
[Sources]
- https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/



Slide 2 of 21

Use the four verbs as the organizing framework for the entire module: Place, Organize, Connect, Secure. These are not Azure product categories; they are a teaching model that connects the platform concepts.

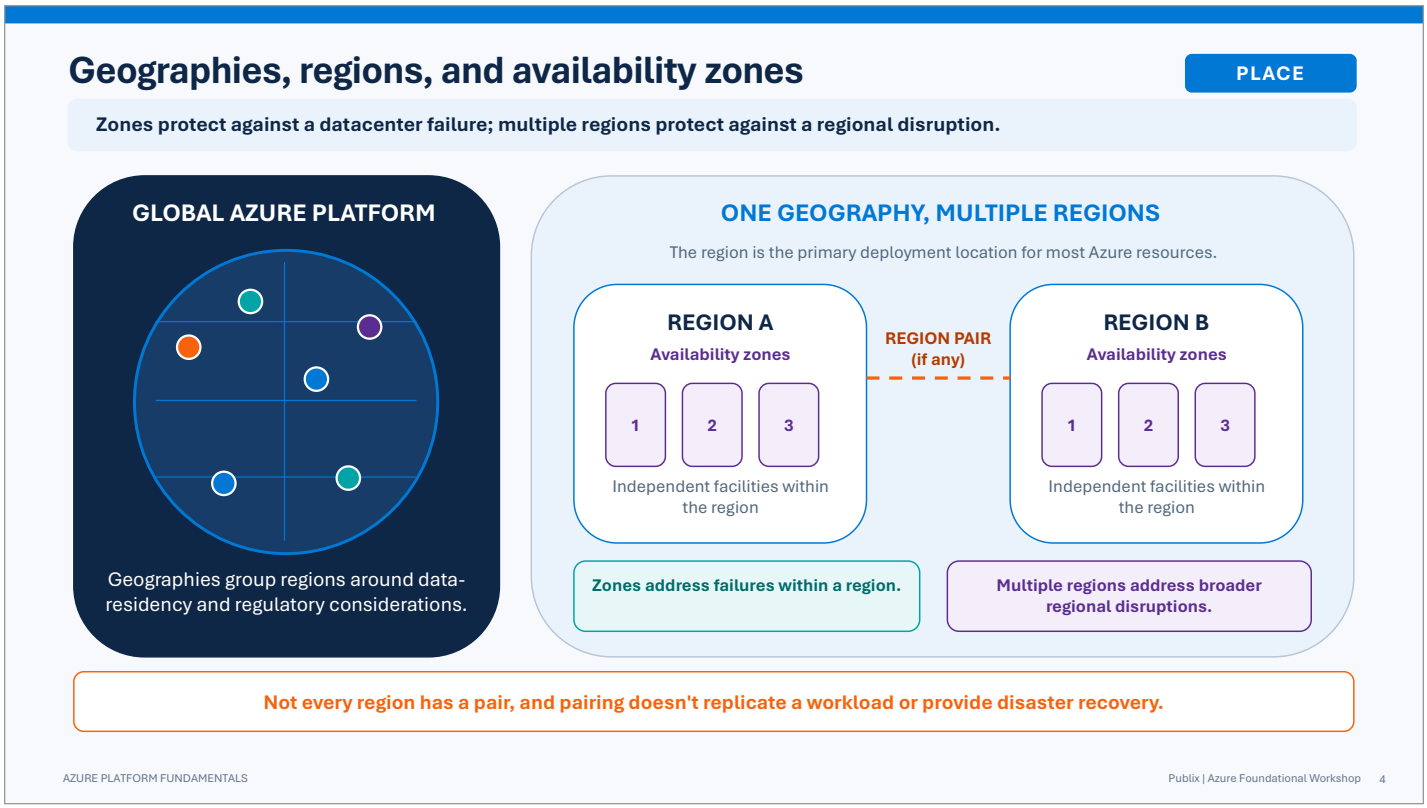
[Sources]
- https://learn.microsoft.com/training/paths/azure-fundamentals-describe-azure-architecture-services/



Slide 3 of 21

Introduce the high-level model first. The remaining slides add implementation detail to each decision. Reinforce that service selection does not replace these foundational decisions.

[Sources]
- https://learn.microsoft.com/azure/architecture/guide/



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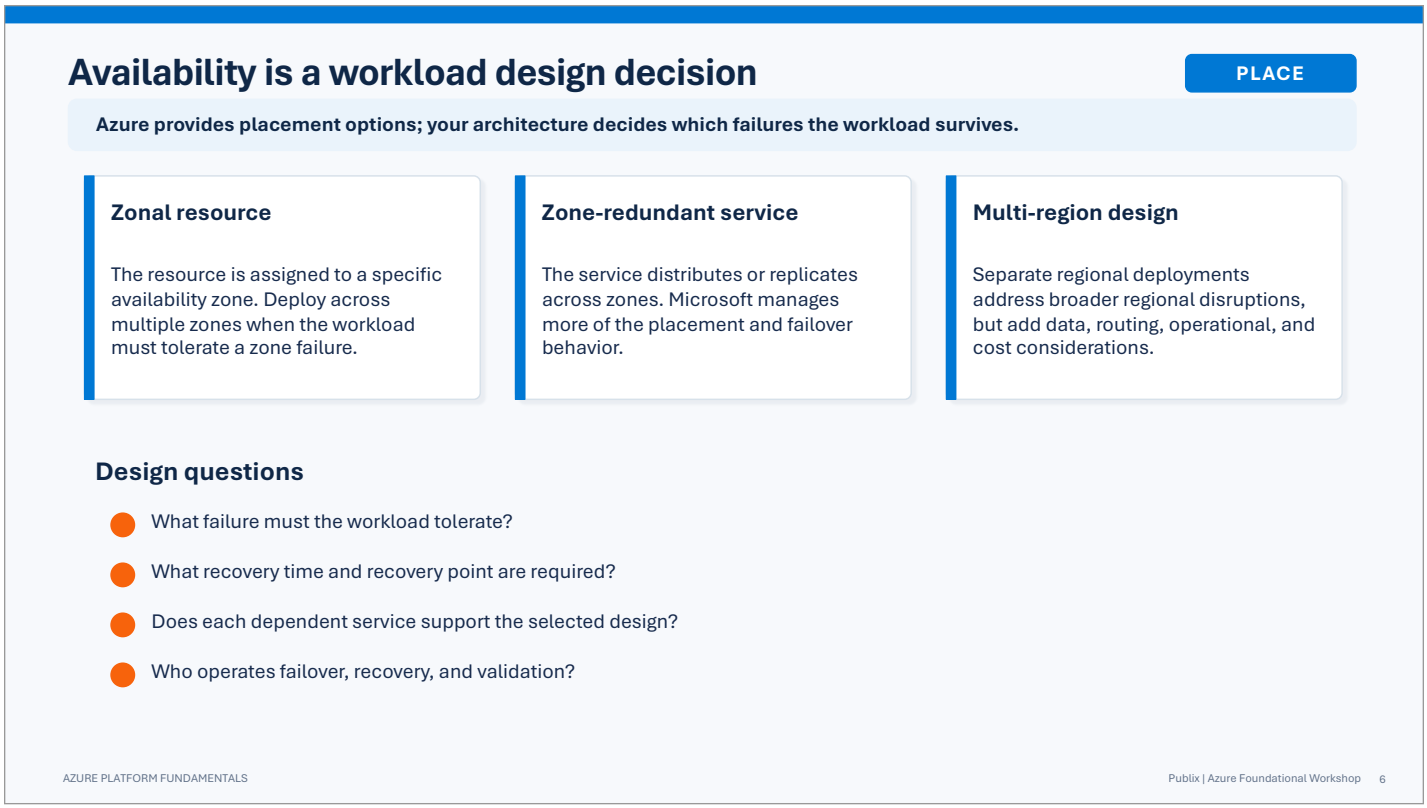
High-level concept: Azure is a global platform organized into geographies and regions. A region is the primary deployment location for most resources. Availability zones provide physically separated locations within a supported region. Some regions are paired with another region, usually in the same geography, and selected Azure services use the relationship for capabilities such as geo-replication, update sequencing, or prioritized recovery. Not every region has a pair. Region pairing does not automatically replicate a workload or provide disaster recovery; workload architecture and service-specific capabilities determine resiliency.

Azure has about 400 data centers and over 70 regions.

[Sources]
- https://learn.microsoft.com/azure/reliability/regions-overview
- https://learn.microsoft.com/azure/reliability/availability-zones-overview
- https://learn.microsoft.com/en-us/azure/reliability/regions-paired

Demonstration

Show portal
Explain marketplace
Show region list



Slide 6 of 21

Avoid implying that selecting a region or zone automatically makes an application highly available. Explain zonal and zone-redundant patterns, then distinguish intra-region availability from multi-region disaster recovery. Reliability requirements should drive the design.

[Sources]
- https://learn.microsoft.com/azure/well-architected/reliability/regions-availability-zones
- https://learn.microsoft.com/azure/reliability/availability-zones-overview



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High-level concept: Azure Resource Manager is Azure's deployment and management service and the primary control plane. Do not describe it as a hosting service. Explain that tools submit management requests to Resource Manager, which interacts with resource providers such as Microsoft.Storage or Microsoft.Web.

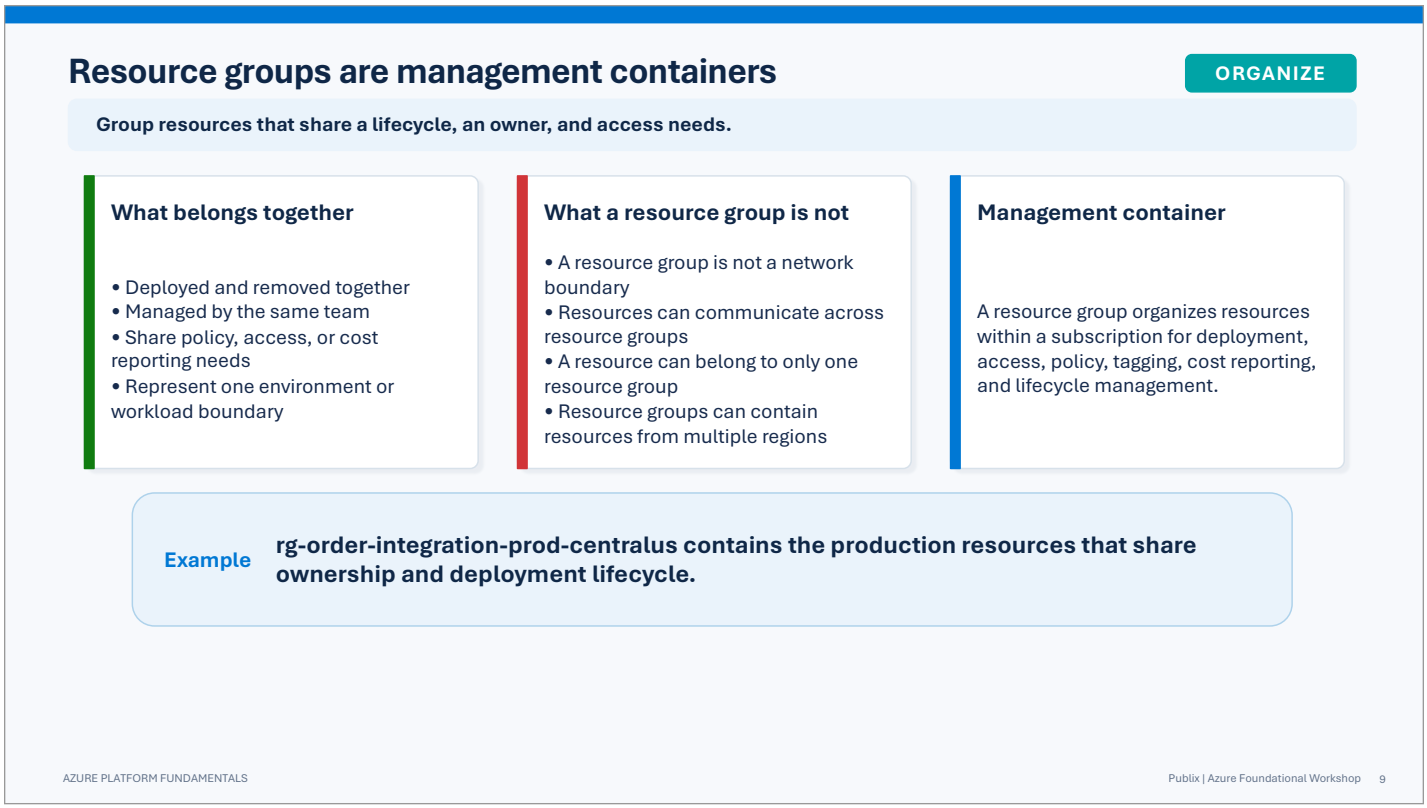
[Sources]
- https://learn.microsoft.com/azure/azure-resource-manager/management/overview
- https://learn.microsoft.com/azure/azure-resource-manager/management/resource-providers-and-types



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Explain hierarchy precisely. The Microsoft Entra tenant provides the identity boundary. Management groups provide governance scope above subscriptions. A subscription is both a management and billing boundary. A resource group is a management container for related resources, not a network boundary. A resource is an individual service instance.

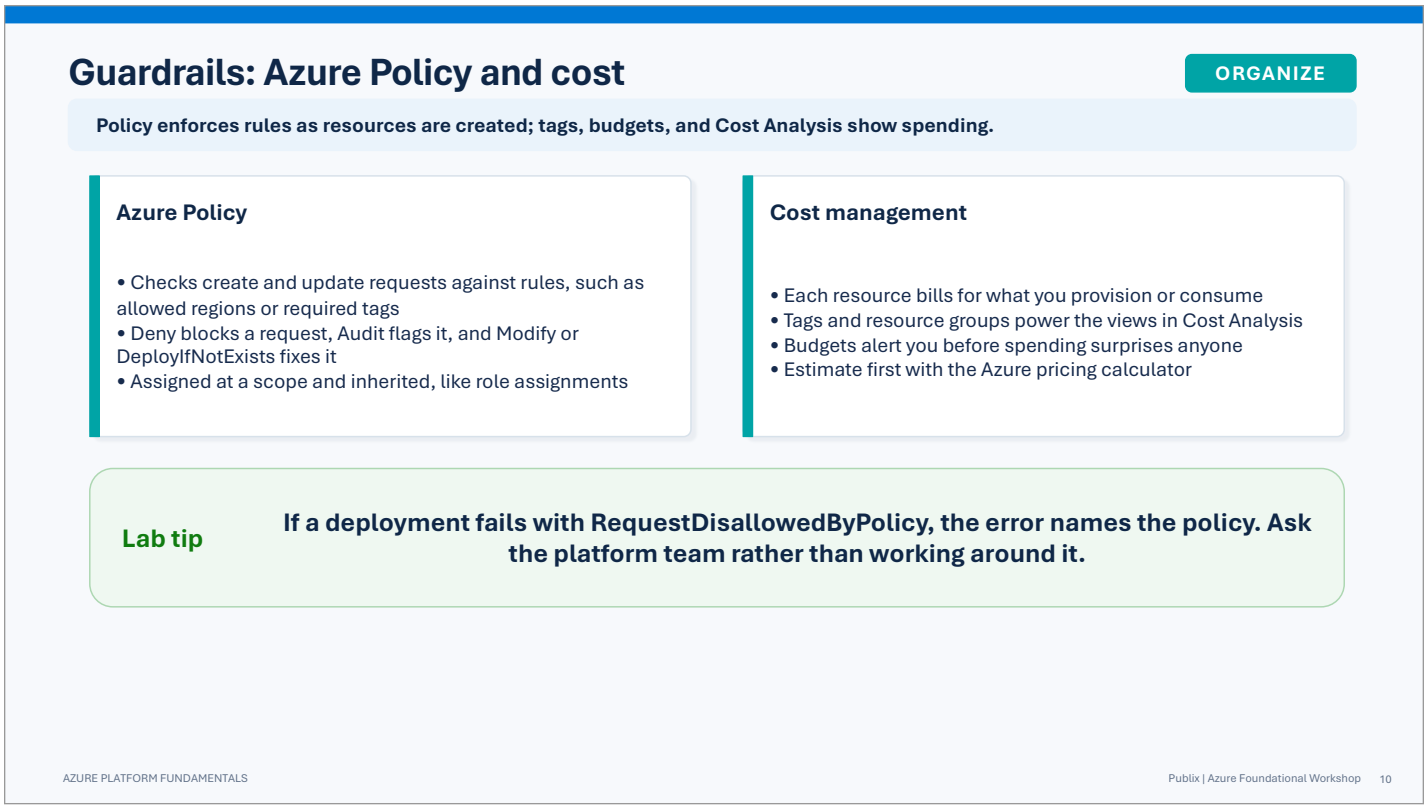
[Sources]
-
https://learn.microsoft.com/azure/cloud-adoption-framework/ready/landing-zone/design-area/resource-org-management-groups
- https://learn.microsoft.com/azure/azure-resource-manager/management/manage-resource-groups-portal



Slide 9 of 21

High-level concept: a resource group is a logical container for Azure resources within a subscription. Use resource groups to organize resources that share management, ownership, policy, access, cost-reporting, deployment, or lifecycle requirements. A resource group is a management scope, not a network or security isolation boundary. Resources in different resource groups can communicate when networking and service configuration permit it.

[Sources]
- https://learn.microsoft.com/azure/azure-resource-manager/management/manage-resource-groups-portal
- https://learn.microsoft.com/azure/azure-resource-manager/management/overview
- https://learn.microsoft.com/azure/cloud-adoption-framework/ready/azure-best-practices/resource-naming



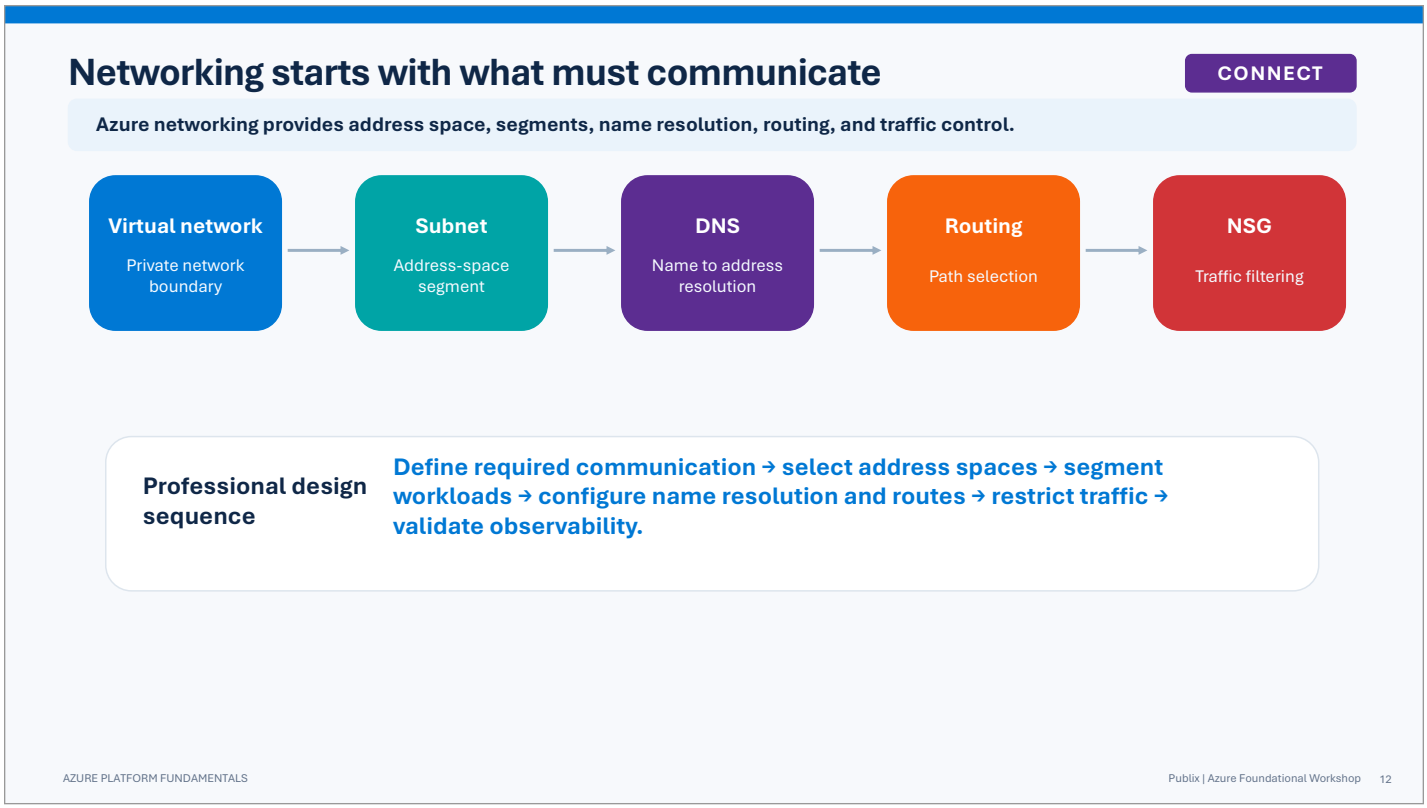
Slide 10 of 21

Two kinds of guardrails attendees will meet on day one. Azure Policy evaluates create and update requests against rules such as allowed regions, required tags, or no public IP addresses. Deny blocks the request, Audit only flags it, and Modify or DeployIfNotExists corrects or adds configuration. Like role assignments, policies are assigned at a scope and inherited. On cost: every resource bills for what it provisions or consumes, so tags and resource groups matter for reporting, budgets give early warning, and the pricing calculator helps before anything is deployed. If a lab deployment fails with RequestDisallowedByPolicy, read the policy name in the error and ask the platform team rather than working around it.

[Sources]
- https://learn.microsoft.com/en-us/azure/governance/policy/overview
- https://learn.microsoft.com/en-us/azure/governance/policy/concepts/effect-basics
- https://learn.microsoft.com/azure/cost-management-billing/costs/quick-acm-cost-analysis
- https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/tutorial-acm-create-budgets
- https://azure.microsoft.com/en-us/pricing/calculator/

Demonstration

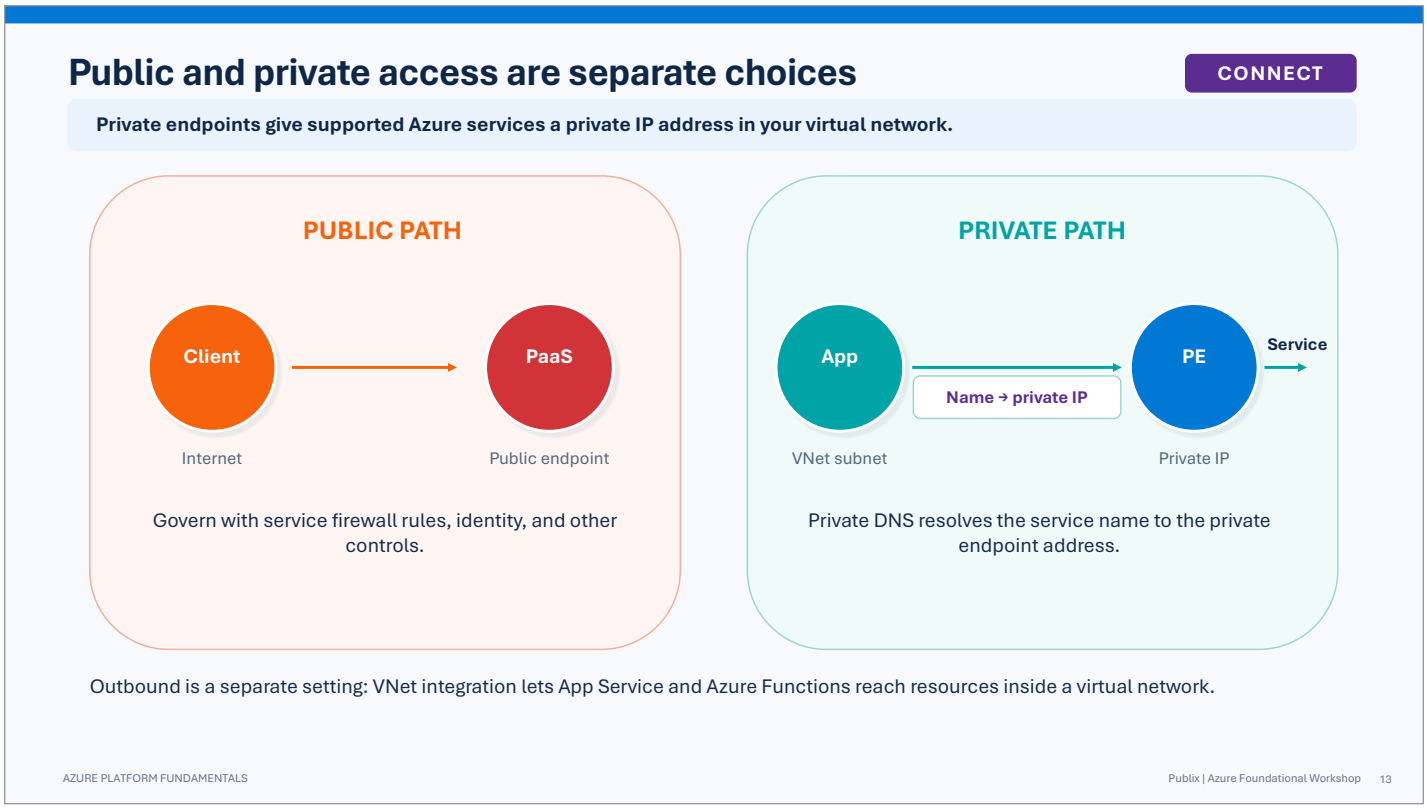
Resource Manager -> All Resources | Subscriptions | Resource Graph Explorer -> Web & Mobile -> App Services
Subscriptions -> Resource providers
Subscriptions -> Cost



Slide 12 of 21

High-level concept: begin with required communication flows, then implement networks and controls. Define each foundational component. Keep this section at L100–200; detailed routing, firewall, and hybrid connectivity are separate topics.

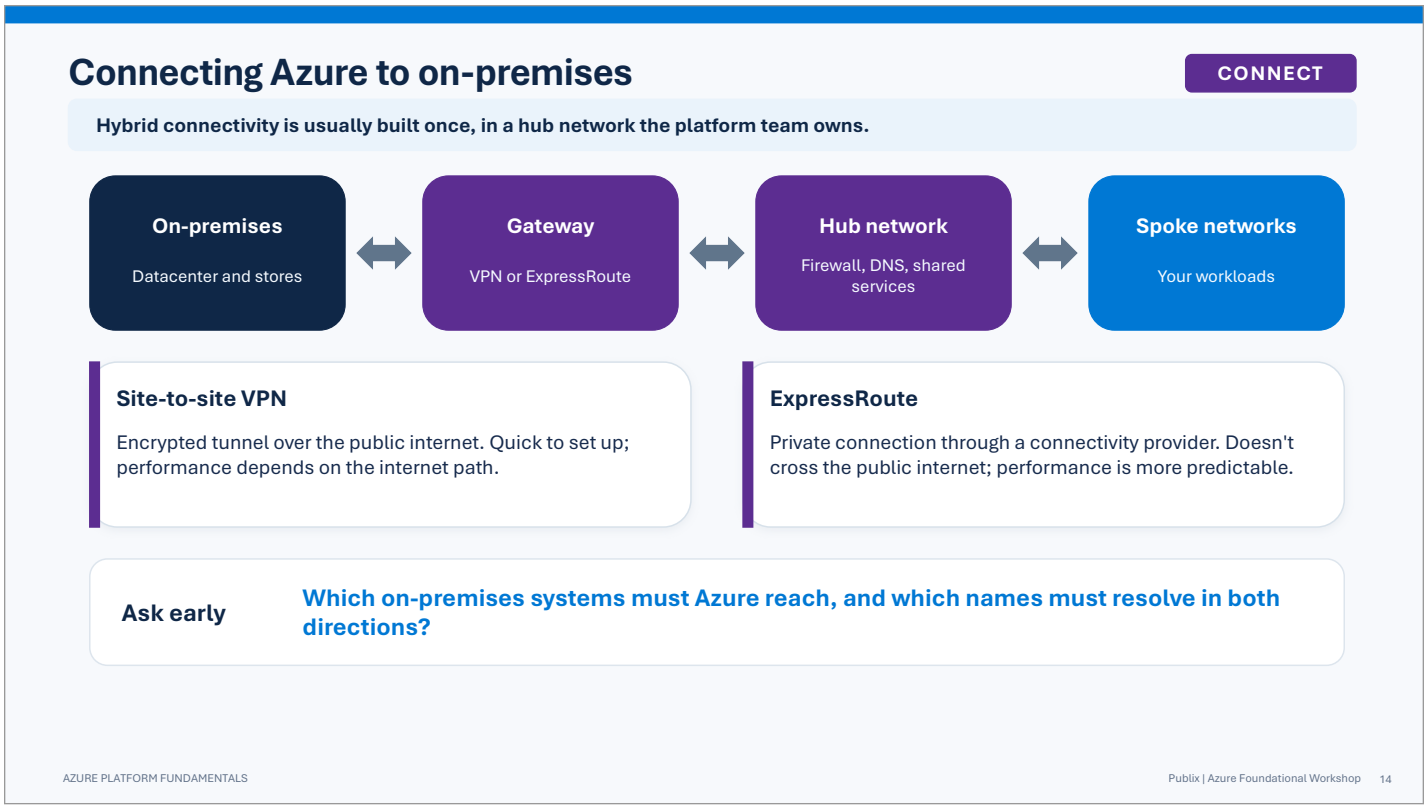
[Sources]
- https://learn.microsoft.com/azure/virtual-network/virtual-networks-overview
- https://learn.microsoft.com/azure/networking/foundations/network-foundations-overview



Slide 13 of 21

Explain that a private endpoint is a network interface with a private IP address for a supported service. DNS is integral because clients normally continue to use the service name. Avoid saying that Private Link alone creates complete end-to-end security; dependent resources and all access paths must also be configured. Private endpoints cover inbound access to a service. Outbound access from App Service or Azure Functions into a virtual network uses VNet integration, which is a separate setting.

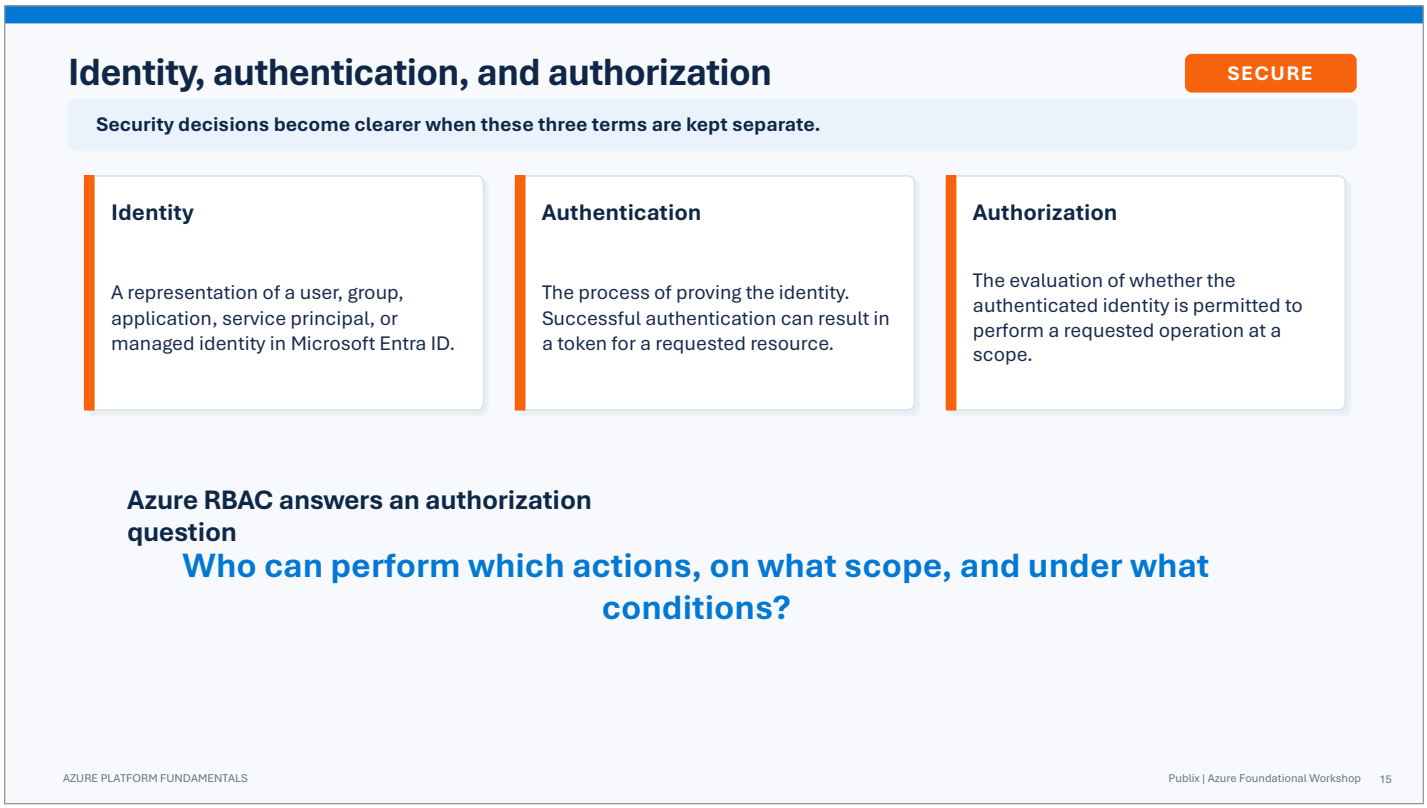
[Sources]
- https://learn.microsoft.com/azure/private-link/private-endpoint-overview
- https://learn.microsoft.com/azure/private-link/private-endpoint-dns
- https://learn.microsoft.com/en-us/azure/app-service/overview-vnet-integration



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Most workloads still need to reach systems in the datacenter or in stores. That connection is usually built once, in a hub virtual network owned by the platform team; workload virtual networks, called spokes, peer to the hub and share its gateway, firewall, and DNS. A site-to-site VPN sends encrypted traffic over the public internet and is quick to set up. ExpressRoute is a private connection through a connectivity provider, with more predictable performance. Name resolution has to work in both directions; Azure DNS Private Resolver can forward queries between Azure and on-premises DNS. Keep this at the concept level; design details belong to the platform team.

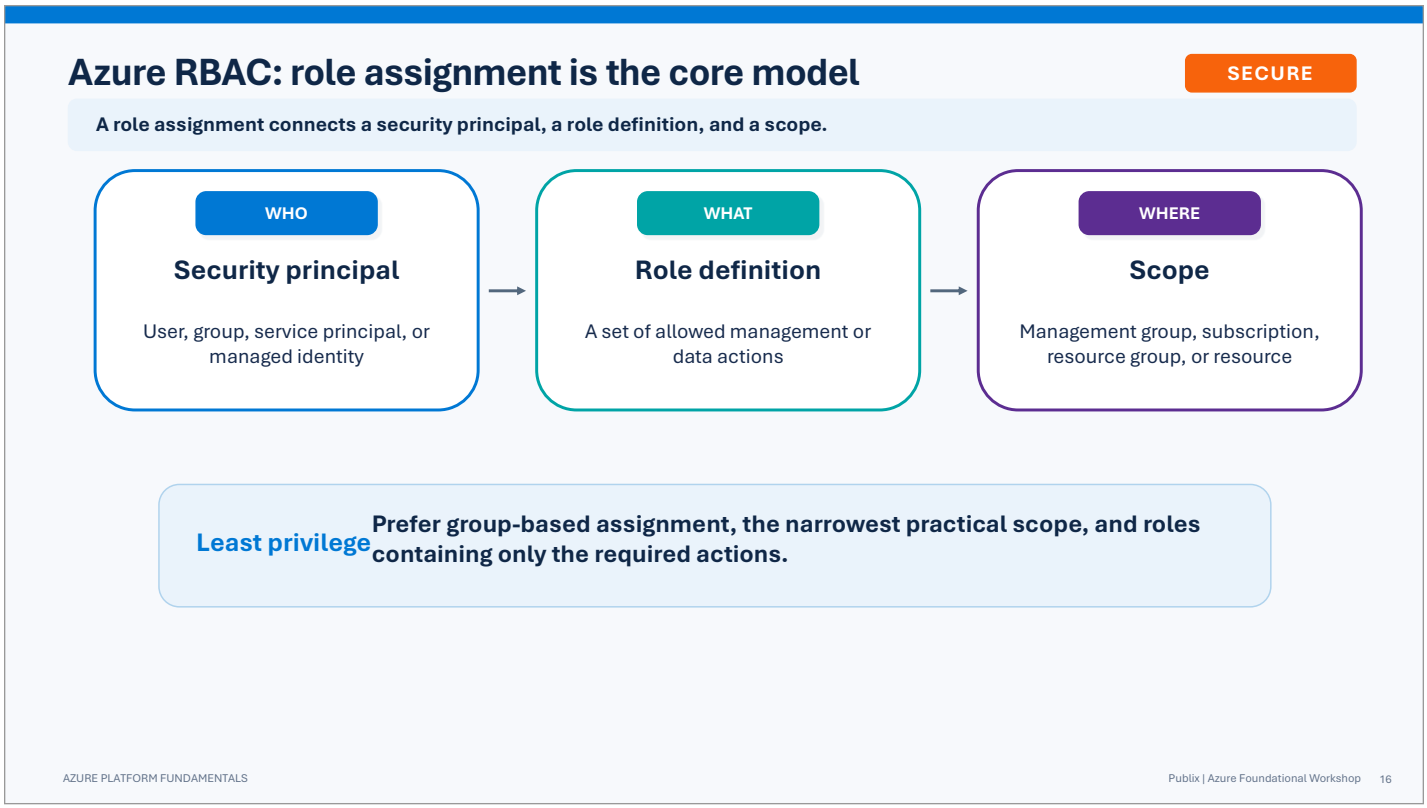
[Sources]
- https://learn.microsoft.com/en-us/azure/vpn-gateway/vpn-gateway-about-vpngateways
- https://learn.microsoft.com/en-us/azure/expressroute/expressroute-introduction
- https://learn.microsoft.com/en-us/azure/architecture/networking/architecture/hub-spoke
- https://learn.microsoft.com/en-us/azure/dns/dns-private-resolver-overview



Slide 15 of 21

Use precise terminology. Microsoft Entra ID is the identity and authentication system. Azure RBAC is an authorization system for Azure resources. Authentication does not imply authorization. A token represents an authenticated security principal and intended audience; the target service still evaluates permissions.

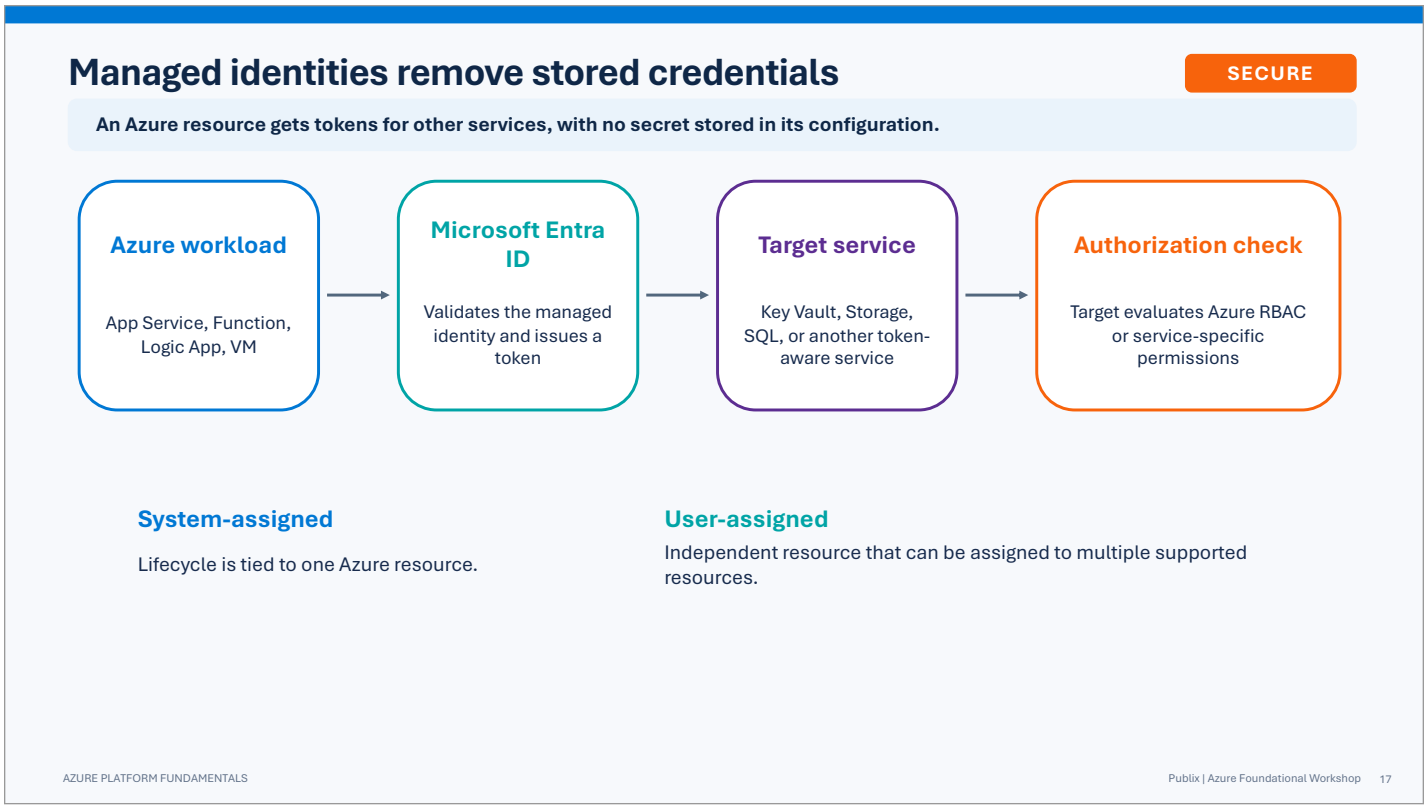
[Sources]
- https://learn.microsoft.com/entra/fundamentals/identity-fundamental-concepts
- https://learn.microsoft.com/azure/role-based-access-control/overview



Slide 16 of 21

Explain role assignments as a three-part relationship. Built-in roles include Owner, Contributor, Reader, and service-specific data roles. Distinguish management-plane roles from data-plane permissions where relevant. Effective access can include inherited assignments.

[Sources]
- https://learn.microsoft.com/azure/role-based-access-control/overview
- https://learn.microsoft.com/azure/role-based-access-control/scope-overview
- https://learn.microsoft.com/azure/role-based-access-control/built-in-roles



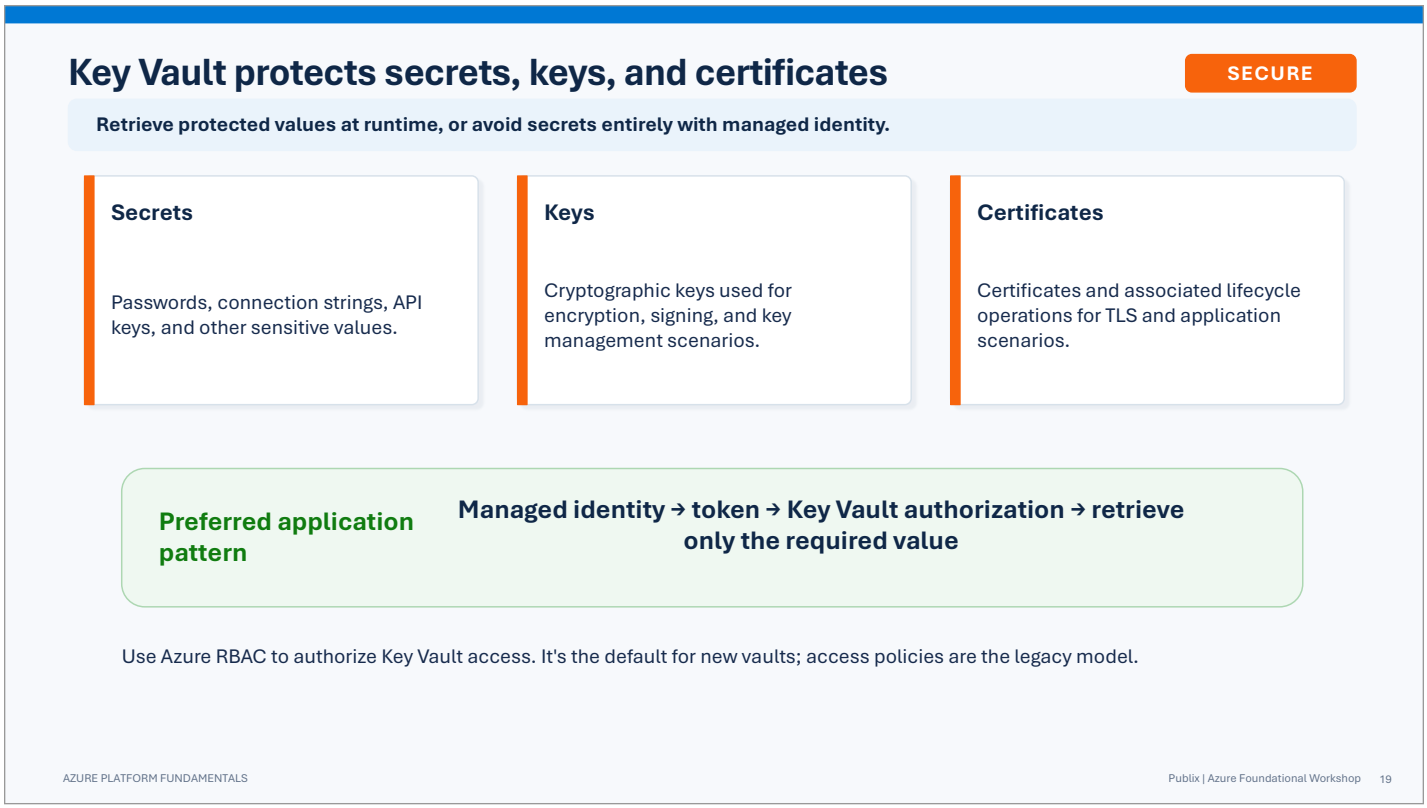
Slide 17 of 21

High-level concept: managed identities address credential management, not authorization. Enabling a managed identity creates an identity, but access still requires an appropriate role assignment or service permission. Explain the lifecycle distinction between system-assigned and user-assigned identities.

[Sources]
- https://learn.microsoft.com/entra/identity/managed-identities-azure-resources/overview
- https://learn.microsoft.com/entra/identity/managed-identities-azure-resources/managed-identity-best-practice-recommendations

Demonstration

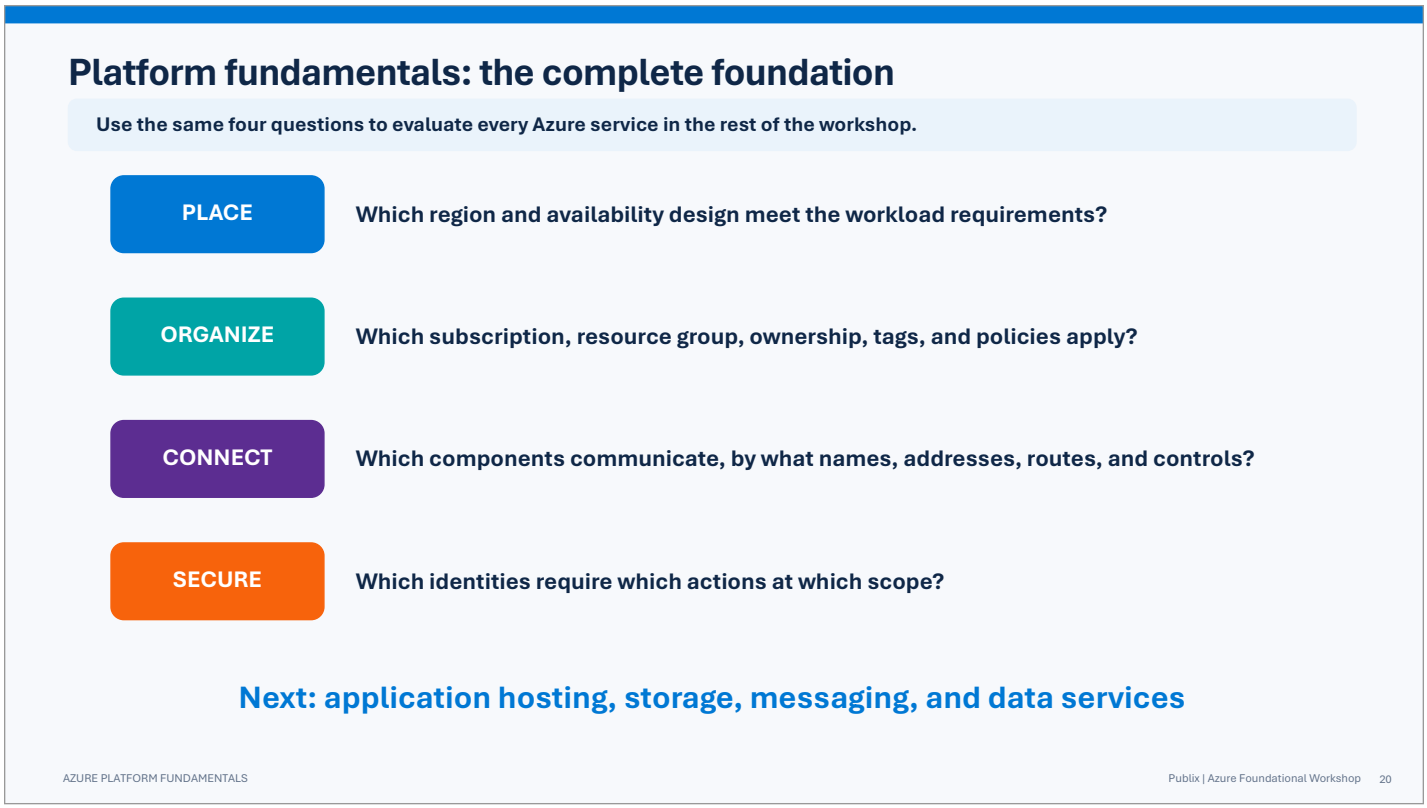
Managed identities, and how to assign priveleges



Slide 19 of 21

Keep Key Vault in the platform fundamentals section because it completes the managed identity story. Explain that Key Vault centralizes protection and access control but does not remove the need to minimize secrets. Prefer token-based service access when available; use Key Vault when protected key material is required. Use Azure RBAC to authorize access to Key Vault: it's the default access model for new vaults, and access policies are the legacy model.

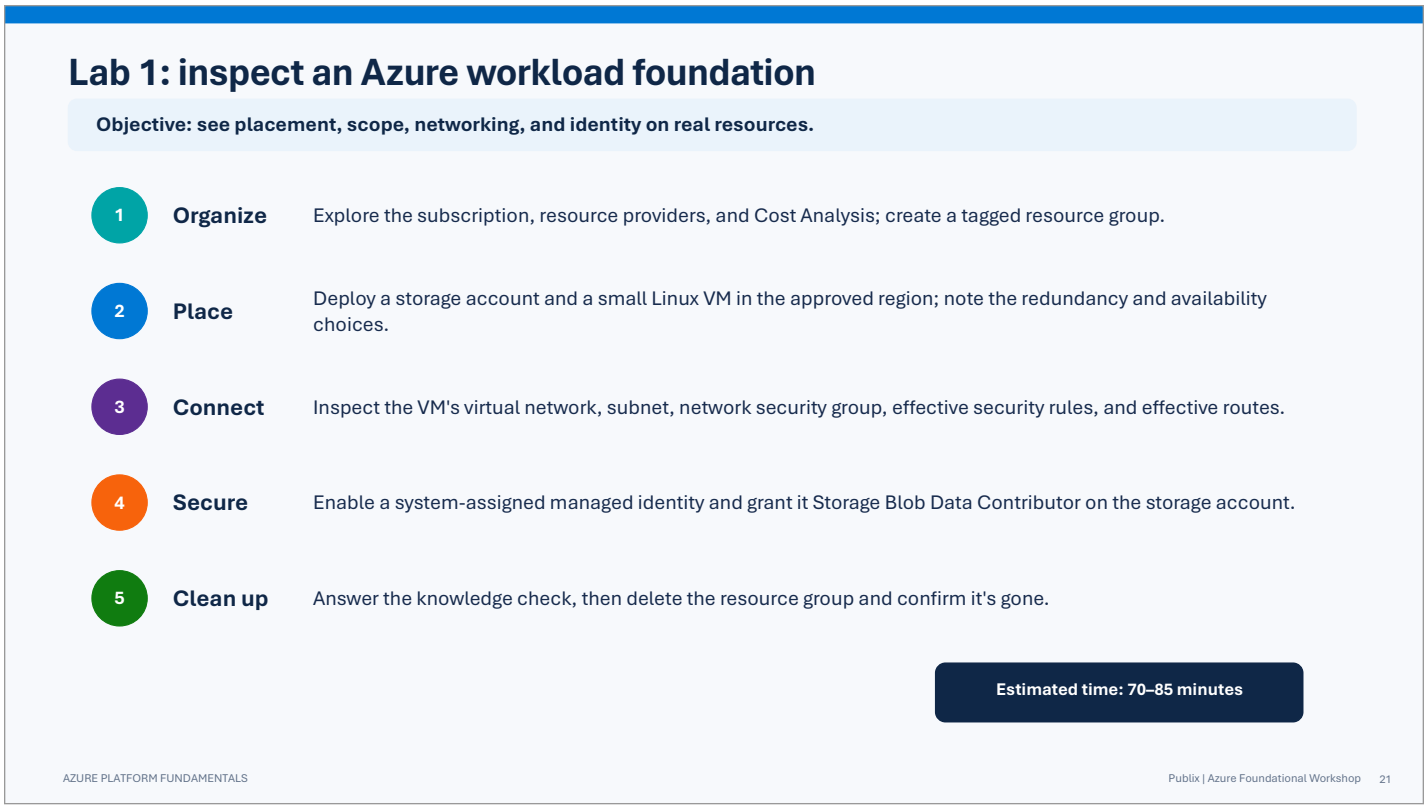
[Sources]
- https://learn.microsoft.com/azure/key-vault/general/overview
- https://learn.microsoft.com/azure/key-vault/general/rbac-guide
- https://learn.microsoft.com/en-us/azure/key-vault/general/access-control-default



Slide 20 of 21

Close by revisiting the four-question model. Invite learners to use it as a checklist when the workshop moves into App Service, Functions, Storage, messaging, and databases.

[Sources]
- https://learn.microsoft.com/azure/architecture/
- https://learn.microsoft.com/azure/well-architected/



Slide 21 of 21

Lab guide: Azure Platform Fundamentals - Lab 1. Attendees work in the Azure portal: review the subscription, resource providers, and Cost Analysis; create a tagged resource group; deploy a storage account and a small Linux VM; inspect the VM's network path, including effective security rules and effective routes; enable a system-assigned managed identity and grant it a data role on the storage account; then delete the resource group. If participants only have read access, run the same checkpoints against precreated resources. Emphasize explanation over portal navigation.

[Sources]
- https://learn.microsoft.com/training/modules/describe-core-architectural-components-of-azure/
- https://learn.microsoft.com/en-us/azure/virtual-network/quickstart-create-virtual-network
- https://learn.microsoft.com/en-us/azure/virtual-network/diagnose-network-traffic-filter-problem
- https://learn.microsoft.com/en-us/entra/identity/managed-identities-azure-resources/grant-managed-identity-resource-access-azure-portal